

1- Transcritical

Bifurcation.

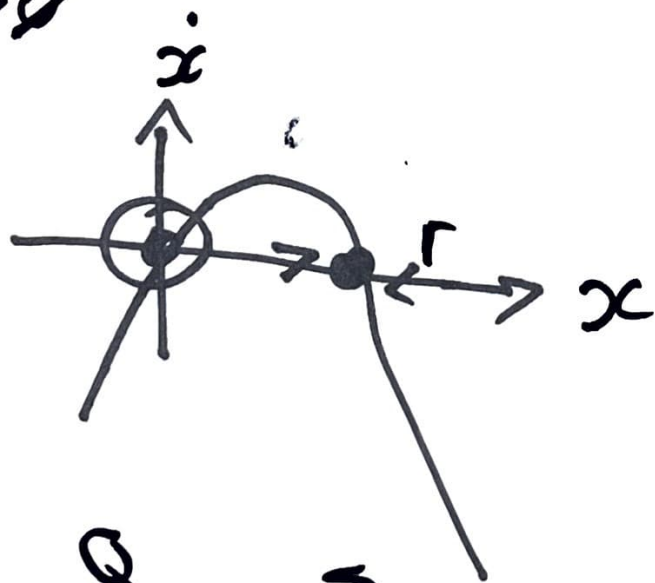
$$\dot{x} = rx - x^2$$

Find fixed points:

$$\dot{x} = x(r - x) \rightarrow x_1^* = 0$$

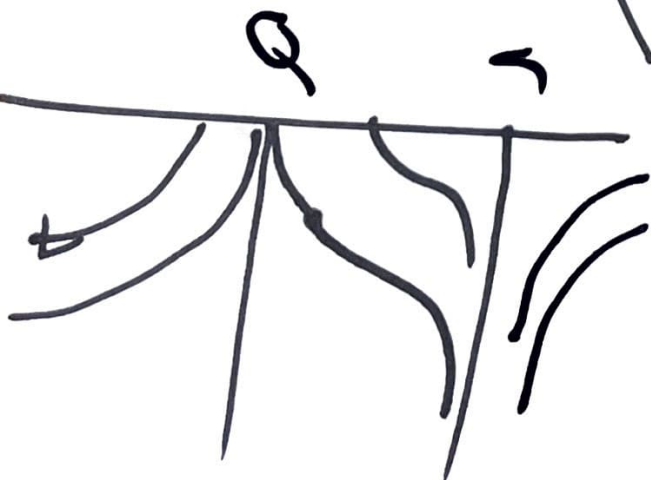
$$x_2^* = r$$

$$r > 0$$



$x_1^* = 0$ unstable

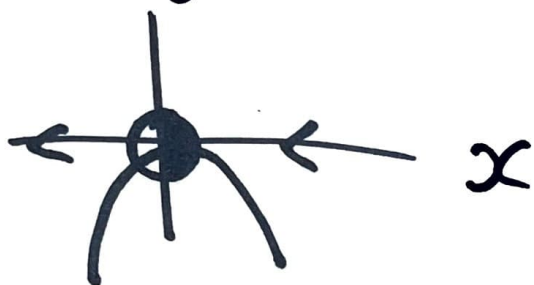
$x_2^* = r$
Stable



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$$r = 0$$

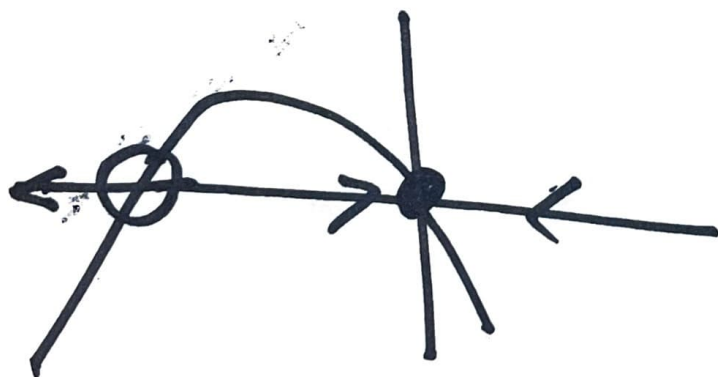
$$\dot{x} = -x^2$$



semi stable
fixed
point

$$r < 0$$

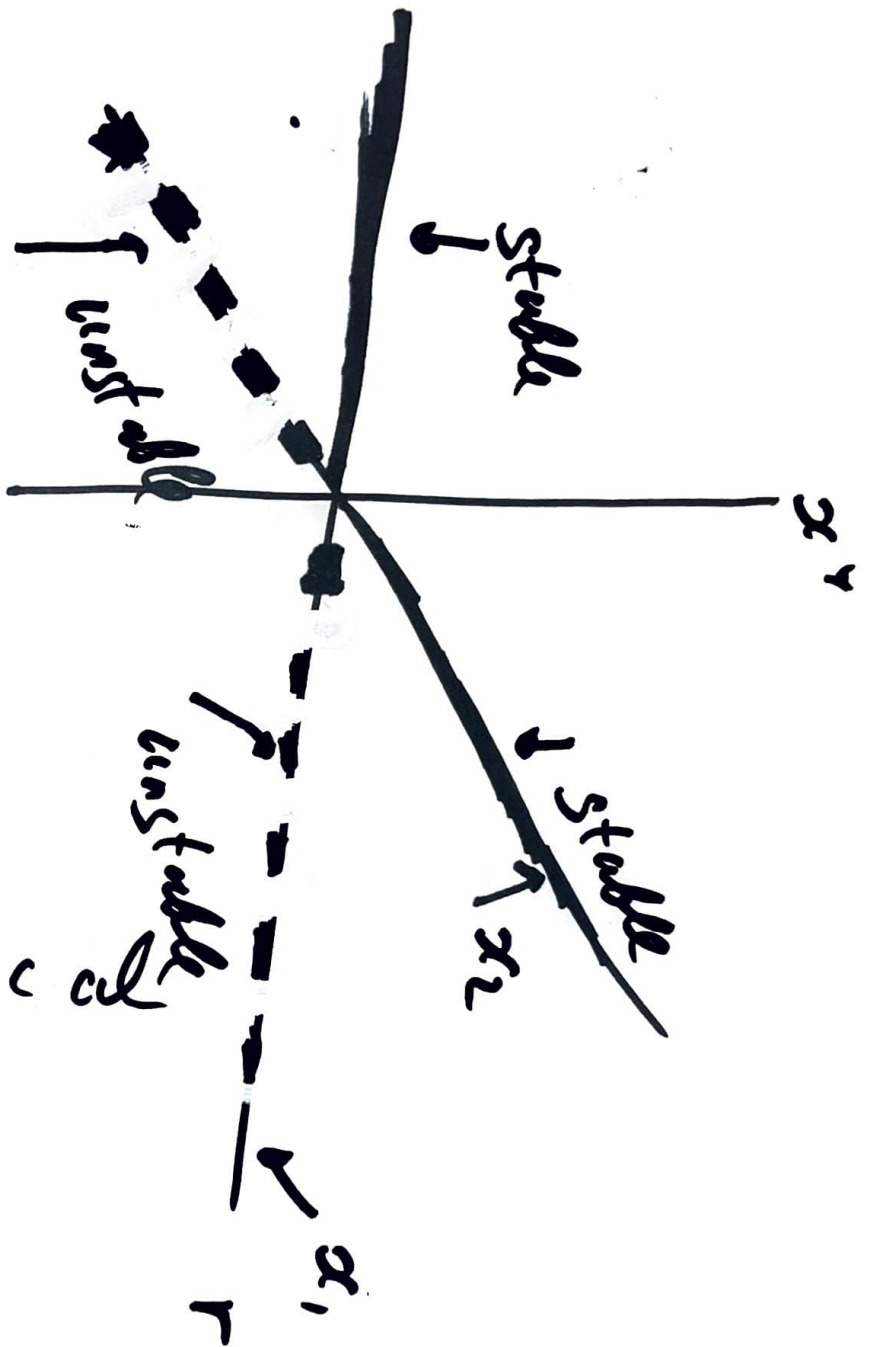
$$\dot{x} = rx - x^2$$



$$x_1^* = 0$$

stable
fixed point

$$x_2^* = r \text{ unstable}$$



Transcritical

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Example

$$\dot{x}(t) = x(1-x^2) - a(1-e^{-bx})$$

Show the system has

transcritical bifurcation,
find a, b curve where

transcritical bifurcation occurs.

Solution

Guess, $x_i^* = 0$, indeed,

$$\dot{x}(t) = f(x(t), a, b)$$

$$f(x, a, b) = x(1-x^2) - a(1-e^{-bx})$$

$$f(x=0, a, b) = 0$$

$$-5' \quad \dot{x} = x(1-x^2) - a(1-e^{-bx})$$

Consider x to be small, $|x| \ll 1$

$$e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots$$

$$e^{-x} \approx 1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \dots$$

$$\dot{x} = x(1-x^2) - a(1 - (1 - bx + \frac{b^2 x^2}{2} - \frac{b^3 x^3}{6} + \dots))$$

$$= x - \cancel{x^3} - a(bx - \frac{b^2 x^2}{2} + \cancel{\frac{b^3 x^3}{6}} + \dots)$$

$$= x(1-ab) + a\frac{b^2 x^2}{2} + \dots$$

Approximation

$$\dot{x} = x(1-ab + a\frac{b^2 x}{2})$$

$$x_1^* = 0,$$

$$x_2^* = 2 \frac{ab-1}{ab^2}$$

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x^*



a

$\leftarrow x^* \neq \emptyset$

b

.

.

.

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Example

$$\dot{x} = r \ln x + x - 1 \quad (*)$$

Show system has

transcritical bifurcation,
change variables to bring
equation (*) to standard
form:

$$\dot{\tilde{x}} = \tilde{x}(r - \tilde{x})$$

Solution

$$x^* = 1;$$

$$\ln(1+u) \approx u - \frac{u^2}{2} \quad |u| < 1$$

$$x = 1+u; \quad \dot{x} = \dot{u}$$

$$\dot{x} = \dot{u} = r \ln(1+u) + 1+u - 1$$

$$\approx r \left(u - \frac{u^2}{2} \right) + u$$

$$= (r+1)u - r \frac{u^2}{2}$$

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$$\dot{u} = (r+1)u - r u^2/2$$

$$u(t) = \frac{2v(t)}{r}$$

$$\dot{u}(t) = \frac{2}{r} \dot{v}(t)$$

$$\begin{aligned} \frac{2}{r} \dot{v} &= (r+1) \cdot \frac{2}{r} v - \frac{r}{2} \cdot \frac{4}{r^2} v^2 \\ \dot{v} &= (r+1)v - v^2 \quad \frac{r}{2} \cdot \frac{4}{r^2} v^2 \\ &= v[r+1-v] \end{aligned}$$

$$r = R-1$$

$$\dot{v} = v[R-v]$$

$$x = 1 + \frac{2}{r} v(t)$$

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$$Li = (r+1)u - \frac{ru^2}{2}$$

$$u = a\gamma; u(t) = a\gamma(t)$$

$$a\dot{\gamma} = (r+1)a\gamma - \frac{r}{2}a^2\gamma^2$$

$$\dot{\gamma} = (r+1)\gamma - \frac{ra}{2}\gamma$$

$$\frac{ra}{2} = 1$$

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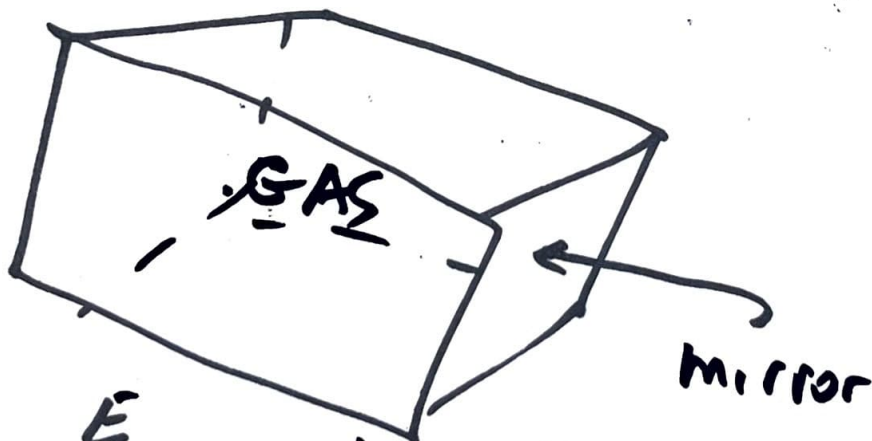
Laser

Light amplification by
Stimulated emission of
radiation.

LABSER

"Active medium" LABSEOR

GAS laser



E
↑ excited state
↑ ground state

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of photons

$n(t)$ - # of photons

$$\frac{d}{dt} n(t) = \text{processes adds the quantity} - \text{processes removing the quantity}$$

$$\frac{d}{dt} n(t) = I_n - \text{Out}$$

↗ gain coefficient

$$I_n = G N(t) n(t)$$

$N(t)$ - # of excited atoms

$$N(t) = N_0 - \lambda n$$

↑
number of
atoms generated

λ - coefficient
of dropping
down

$$\text{Out} = \beta n, \quad \beta - \text{rate of photon dissipation}$$

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$$\dot{n} = G n N - R n \quad (1)$$

$$= G n (N_0 - d n) - R n \quad (2)$$

$$= n (G N_0 - R) - d G n^2 \quad (3)$$

$$= n [G N_0 - R - d G n] \quad (4)$$

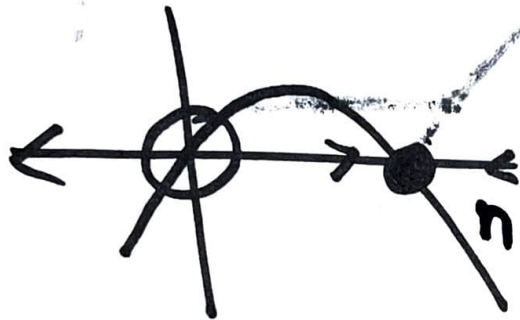
$n(t) = 0$ - fixed point

$$n(t) = \frac{G N_0 - R}{d G} = \frac{N_0}{d} - \frac{R}{d G}$$

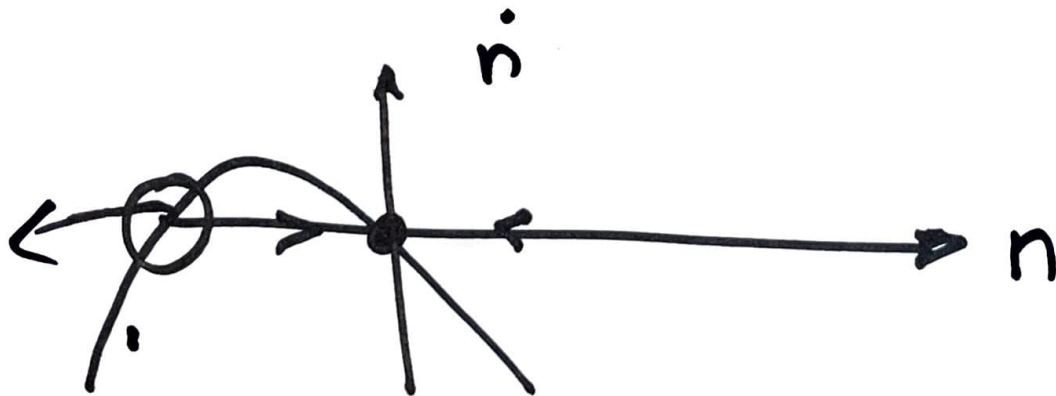
$$N_0 > R/G, \quad N_0 < R/G$$

$$12 \quad N_0 > R/\sigma$$

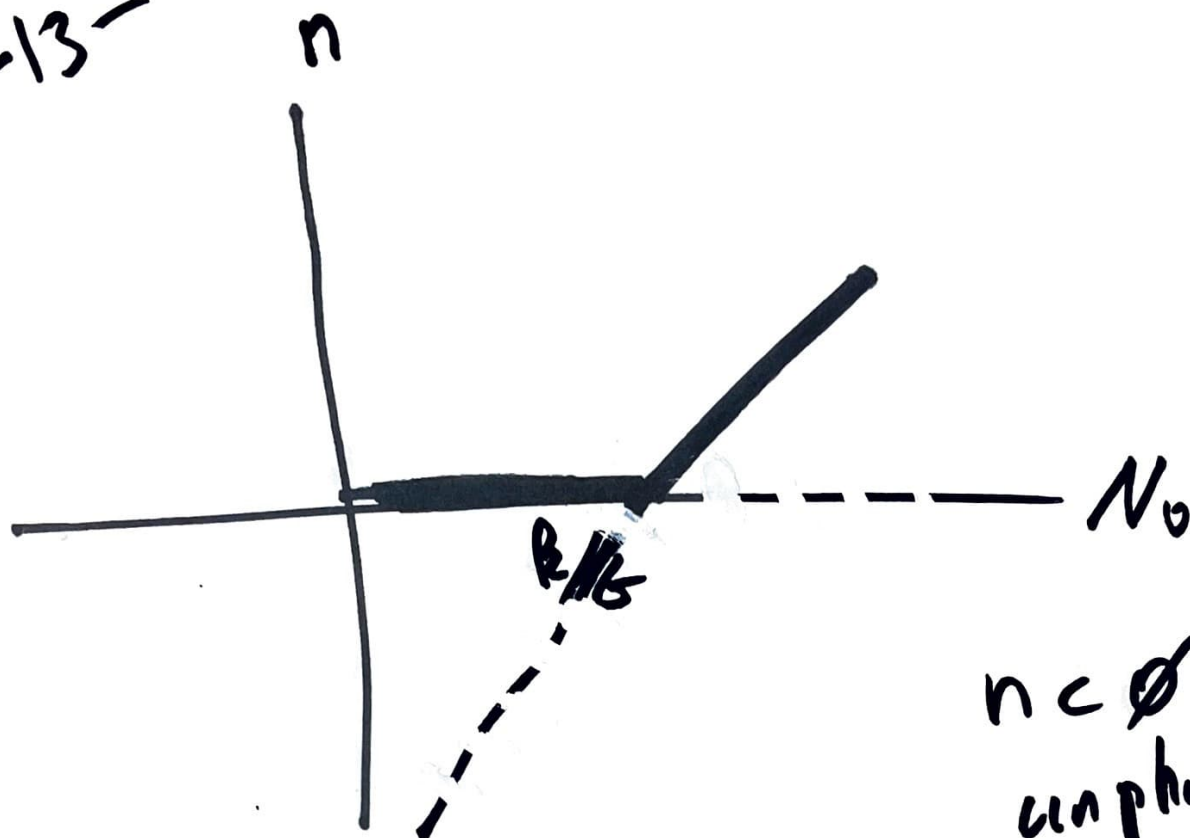
$$\dot{n} = n [G N_0 - R - \alpha G n]$$



$$N_0 < R/\sigma$$



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$n < \emptyset$
unphysical

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Pitchfork

bifurcation

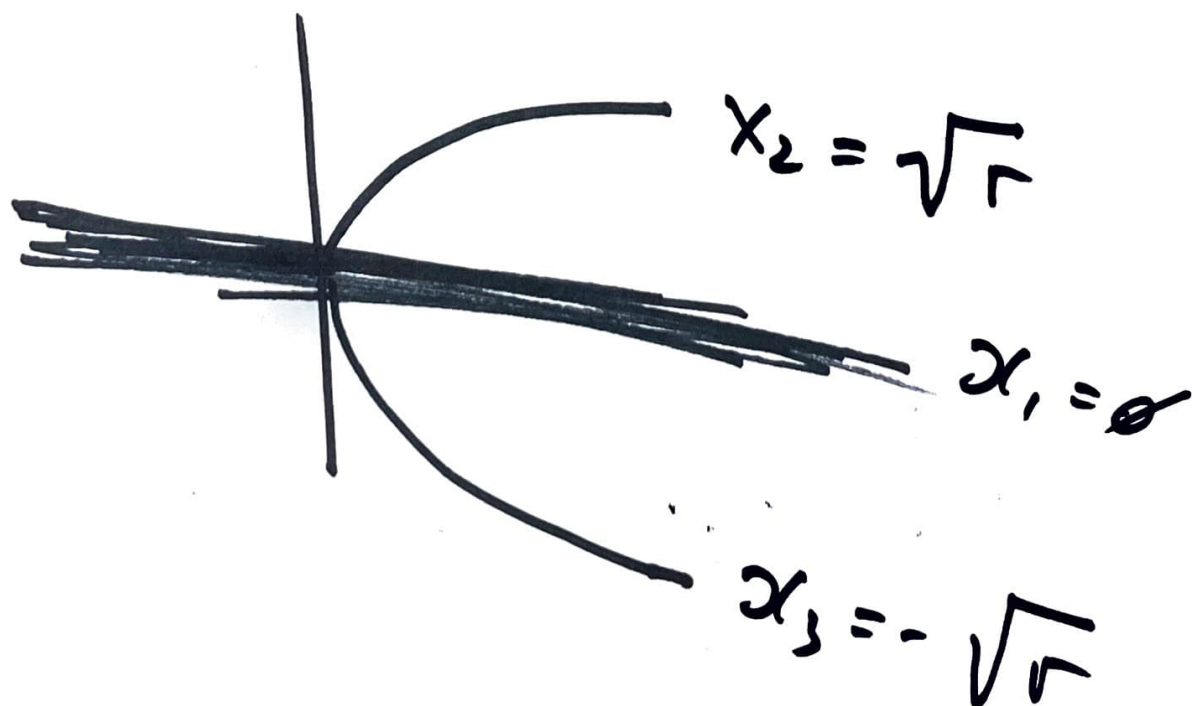
$$\dot{x} = rx - x^3$$

Supercritical

$$\dot{x} = x(r - x^2)$$

$$x_1^* = 0$$

$$x_{2,3}^* = \pm \sqrt{r}$$



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$$\dot{x} = f(x, r)$$

$$f(x, r) = rx - x^3$$

$$\frac{\partial f(x, r)}{\partial x}$$

$$= r - 3x^2$$

$$x = 0: \quad \begin{array}{ll} r > 0 & , \text{ unstable} \\ r < 0 & \text{ stable} \end{array}$$

$$x = \pm\sqrt{r}, \quad \underline{r > 0}$$

$$\frac{\partial f}{\partial x} = r - 3|r| = -2r$$

stable